

Installation and Configuration of phpMyAdmin in the Apache2 LXC (MariaDB in a separate Container)

Introduction to phpMyAdmin

phpMyAdmin is a web-based administration interface for MySQL and MariaDB databases. It allows users to easily create, manage, and edit databases via a browser without having to use the command line directly.

Typical Use Cases for phpMyAdmin include:

- **Database Administration:** Creating, deleting, and editing databases and tables.
- **Data Management:** Inserting, editing, or deleting data records.
- **SQL Queries:** Executing SQL commands and queries directly via the user interface.
- **Backup and Recovery:** Exporting and importing databases for backups or migrations.
- **User and Permissions Management:** Creating database users and assigning permissions.

phpMyAdmin is primarily used to simplify database management for administrators and developers and is especially common in web server environments.

Prerequisites

- Apache2 is installed and running in the container.
 - MariaDB is running in a separate LXC container (**mariadb**) and is configured to accept external connections (e.g., **bind-address = 192.168.137.120**) and has an Admin user.
 - The containers are located in the same network.
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1. Install phpMyAdmin in the Web Server Container

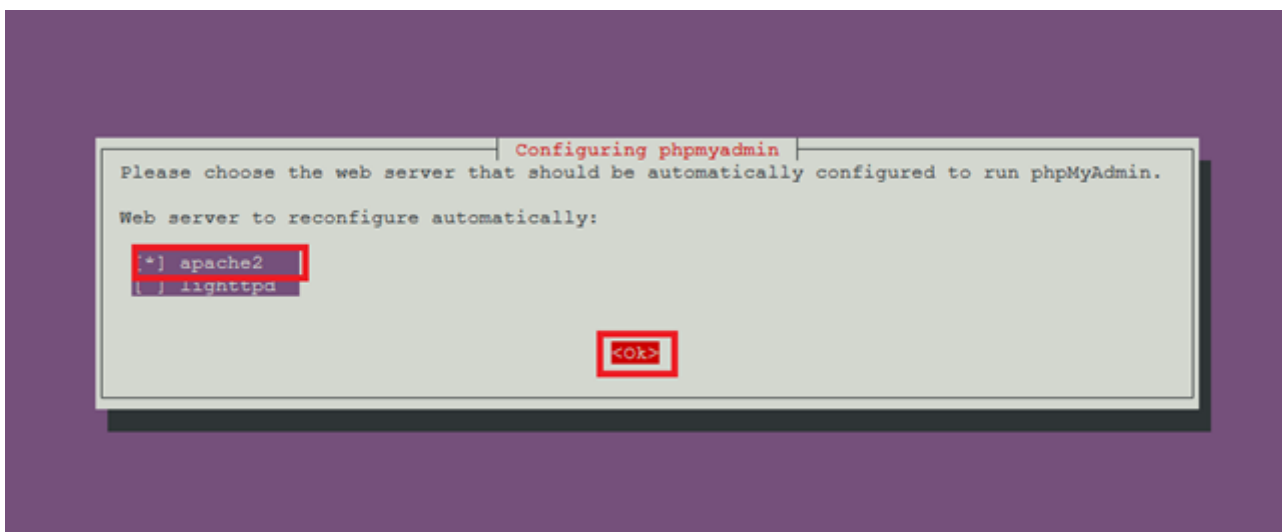
 Note: "phpMyAdmin" is a **PHP web application** that runs on an existing web server (e.g., Apache or Nginx). Therefore, phpMyAdmin cannot be started "alone" in a container—it always requires a web server and an existing database connection to function.

```
apt update
apt install -y phpmyadmin php-mysql
```

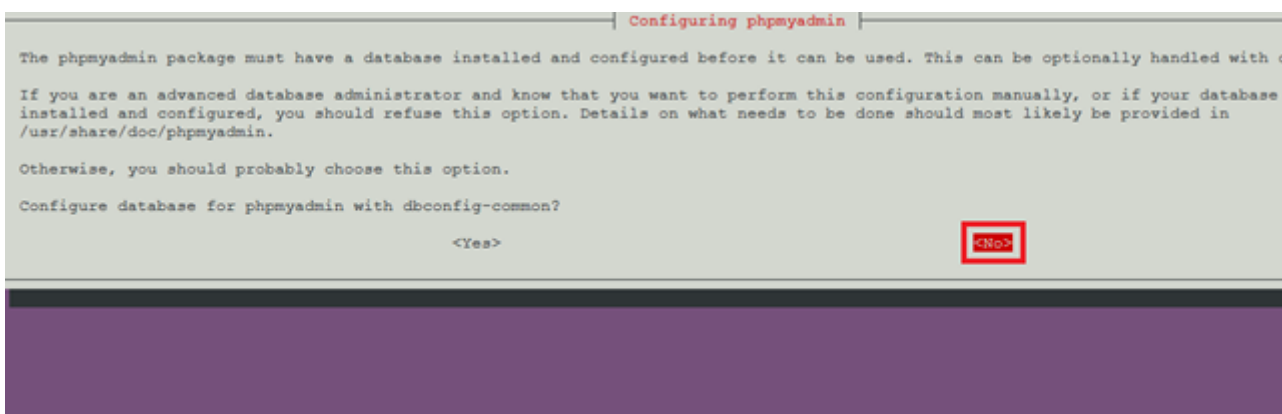
```
pdal@apache101:~$ sudo apt update
[sudo] password for pdal:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1170 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [246 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1279 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [272 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1092 kB]
Fetched 4311 kB in 3s (1515 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
pdal@apache101:~$ sudo apt install phpmyadmin php-mysql
```

💡 During Installation:

Web server selection: When prompted, select **apache2** (Spacebar → Tab → OK).



Database configuration with dbconfig-common: → **No**, as MariaDB is running externally.



If the web server selection does not appear, manually link phpMyAdmin:

```
ln -s /usr/share/phpmyadmin /var/www/html/phpmyadmin
```

Afterward, restart Apache:

```
systemctl restart apache2
```

⚙️ 2. Configure Remote MariaDB Connection in phpMyAdmin

Since we are running MariaDB in a separate container, phpMyAdmin needs to know that the database server is not **localhost** but the **external IP** of the MariaDB container.

Edit the configuration file **config-db.php**, as this contains the primary server variables used by **config.inc.php**:

```
sudo nano /etc/phpmyadmin/conf.d/config-db.php
```

In this file, find the line defining the server and enter the IP address of the MariaDB container (e.g., **192.168.137.120**).

Attention: The content of this file may vary. Typically, the relevant part for defining the database host looks like this:

```
// Add this line to the file or change the existing host definition:  
$dbserver = '192.168.137.120'; // IP address of the MariaDB container
```

🔗 **Important Change:** The phpMyAdmin web server now connects via TCP to the specified IP. Since **\$dbserver** is no longer **'localhost'**, **connect_type = 'tcp'** is automatically set in **config.inc.php**.

Save and close (**Ctrl + O** -> **Enter** -> **Ctrl + X**).

🔄 3. Restart Apache

```
systemctl restart apache2
```

🌐 4. Access phpMyAdmin in the Browser

Open in your browser:

http://IP-of-the-phpMyAdmin-Container/phpmyadmin

Example:

http://192.168.137.101/phpmyadmin

Log in with the MySQL/MariaDB user that has access from the phpMyAdmin container.



5. Troubleshooting

Note: this error should only occur if you have not fully executed the script "Installation and Configuration of MariaDB in the LXC Container." If you have correctly created the user, check connectivity to the database LXC.

Access denied

Switch to the MariaDB container and ensure that the user in MariaDB is correctly granted access for our local network:

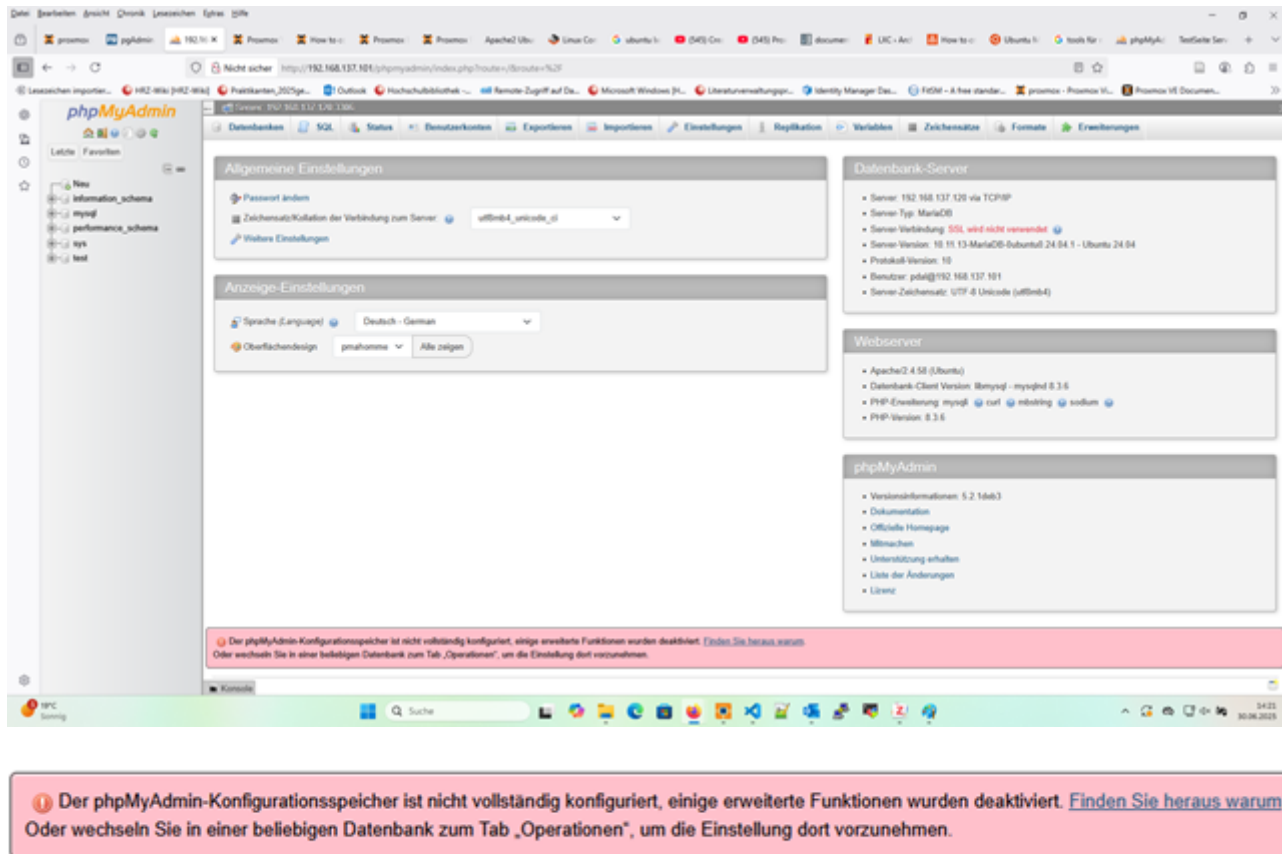
```
CREATE USER 'pdal'@'192.168.137.%' IDENTIFIED BY 'JadeHS20';  
GRANT ALL PRIVILEGES ON *.* TO 'pdal'@'192.168.137.%' WITH GRANT OPTION;  
FLUSH PRIVILEGES;
```

Note: If you only want to allow access from one LXC container, change the last octet of the IP address from 'pdal'@'192.168.137.%' to 'pdal'@'192.168.137.123' (IP of the container from which access should be granted). This would significantly increase security.

☑ Conclusion

phpMyAdmin is now ready and connects to the external MariaDB database. You can manage databases, create users, make backups, etc., via the web interface.

Currently, we see a notice in the phpMyAdmin WebGUI stating that the configuration storage is not fully configured. This is explained in the following steps.



6. Setting up the phpMyAdmin Configuration Storage (Advanced Features)

Set up the internal phpMyAdmin configuration storage to use advanced features such as Designer, Bookmarks, and Relation View.

💡 Explanation:

phpMyAdmin requires certain internal tables in the database to support features like Bookmarks, Relations, or PDF export. These tables are defined by the `create_tables.sql` file. Since **phpMyAdmin is running in the Apache container** in our environment, but **MariaDB is in a separate container**, phpMyAdmin cannot **directly create the tables itself**.

We therefore choose the secure and simple path via the web interface:

- Go to the **"Databases"** tab.
- Under "Create new database," enter the name **phpmyadmin**. Click "Create."
- You are automatically taken to the newly created database; otherwise, select the **phpmyadmin** database.
- Go to the **"More"** tab (or **"Operations"** tab, depending on the version).

- At the top, there is a message: "The phpMyAdmin configuration storage is not completely configured, some extended features have been deactivated." Click on the link "**Find out why.**"
- Now click on "**Create**" and the tables will be created automatically.

The configuration is complete.

In Summary:

- This step initializes the phpMyAdmin-specific tables in the database so that the web interface can be used correctly and completely.

Optional Task: Set up an Alias for PhpMyAdmin

The document "Apache2 Webserver & User Management in the LXC Container" explains how to set up an alias.

Move the phpMyAdmin directory from `/var/www/html/phpmyadmin` to `/var/www/phpmyadmin` and set up an **Alias** for phpMyAdmin. This keeps the HTML directory free for your applications.

Sources

- "Introduction — phpMyAdmin 6.0.0-dev documentation". Accessed: September 25, 2025. [Online]. Available at: [Einführung](#)
 - "Requirements — phpMyAdmin 6.0.0-dev documentation". Accessed: September 25, 2025. [Online]. Available at: [Anforderungen](#)
 - "Installation — phpMyAdmin 6.0.0-dev documentation". Accessed: September 25, 2025. [Online]. Available at: [Installation](#)
 - "Configuration — phpMyAdmin 6.0.0-dev documentation". Accessed: September 25, 2025. [Online]. Available at: [Konfiguration](#)
 - "User Guide — phpMyAdmin 6.0.0-dev documentation". Accessed: September 25, 2025. [Online]. Available at: [Benutzerhandfbuch](#)
 - "FAQ - Frequently Asked Questions — phpMyAdmin 6.0.0-dev documentation". Accessed: September 25, 2025. [Online]. Available at: [FAQ](#)
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